STATION 5: MY NUMBER

N=266

1.) Student no.: 26118266
N=266

2.) BINARY

DIVISION	QUOTIENT	REMAINDER
266/2	133	0
133/2	66	1
66/2	33	0
33/2	16	1
16/2	8	0
8/2	4	0
4/2	2	0
2/2	1	0
1/2	0	1

OCTAL

DIVISION	QUOTIENT	REMAINDER
266/8	33	2
33/8	4	1
4/8	0	4

HEXADECIMAL

DIVISION	QUOTIENT	REMAINDER	HEX DIGIT
266/16	16	10	A
16/16	1	0	0
1/16	0	1	1

3.) BINARY VERIFICATION

PLACE VALUE	2^8 (256)	2^7 (128)	2^6 (64)	2^5 (32)	2^4 (16)	2^3 (8)	2^2 (4)	2^1 (2)	2^0 (1)
DIGIT	1	0	0	0	0	1	0	1	0

Calculation: (1*256) + (1*8) + (1*2) = 256+8+2 = 266_{10}

OCTAL VERIFICATION

PLACE VALUE	8^2 (64)	8^1 (8)	8^0 (1)
DIGIT	4	1	2

Calculation: (4*64) + (1*8) + (2*1) = 256+8+2 = 266_{10}

HEXADECIMAL VERIFICATION

PLACE VALUE	16^2 (256)	16^1 (16)	16^0 (1)
DIGIT	1	0	A

Calculation: (1*256) + (0*16) + (10*1) = 256+10 = 266_{10}

4.) BINARY TO OCTAL

100 - 4

001 - 1 $=412_8$

010 - 2

BINARY TO HEXADECIMAL

0001 - 1

0000 - 0 $=10A_{16}$

1010 - A

5.) My classmate made a mistake of reading the remainders from top to bottom instead of bottom to top. Resulting to writing the sequence backwards, making it 011101_2 (29_{10}).

The Correct Answer is: 101110_2